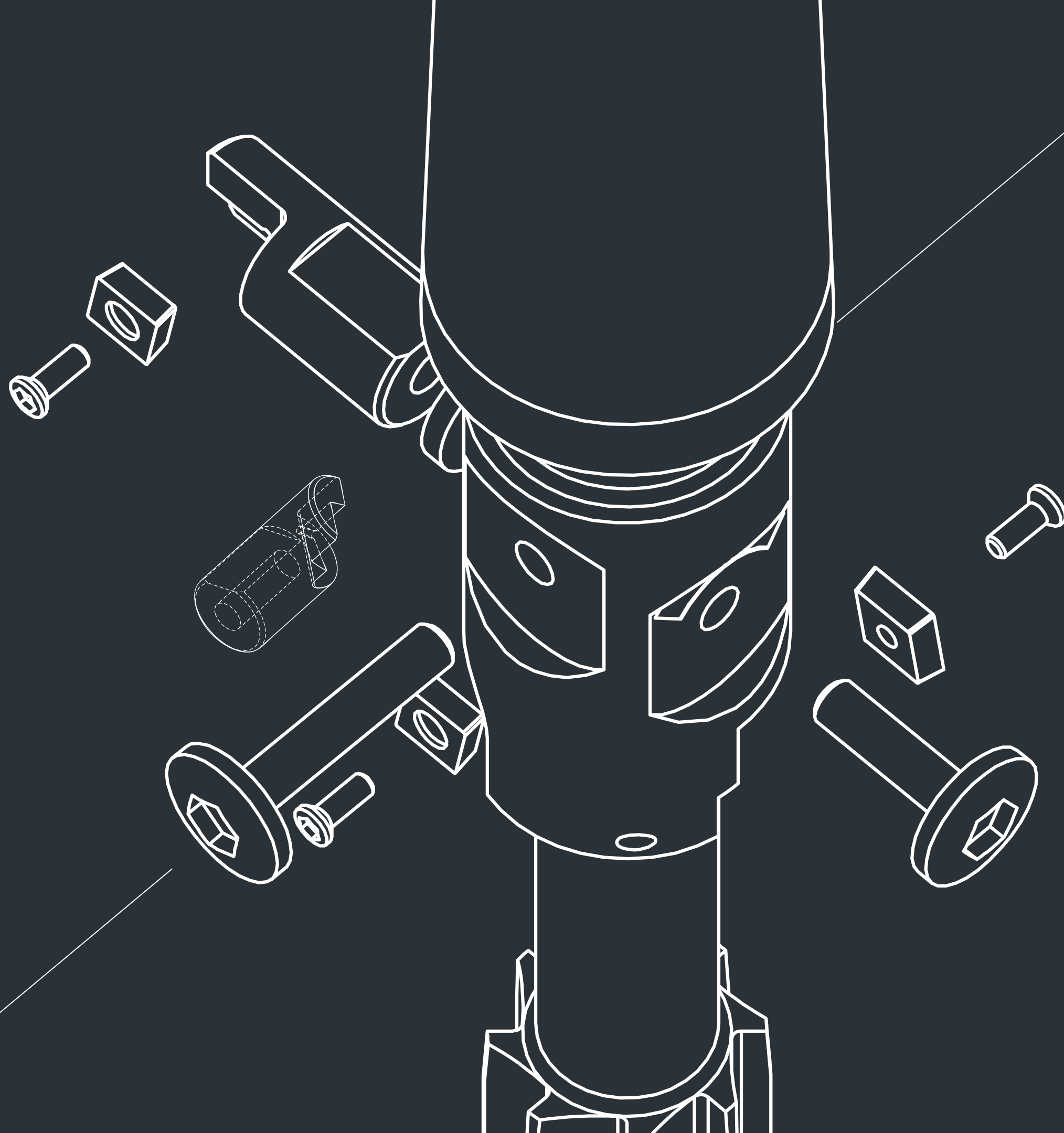




CropSentry Research Presentation

The Problem

Industry Concerns

Over \$60B is spent annually on crop protection applied on a calendar schedule or after symptoms appear, not when stress begins.



\$220B

Lost from plant disease

20-40%

Globally lost plants due to diseases

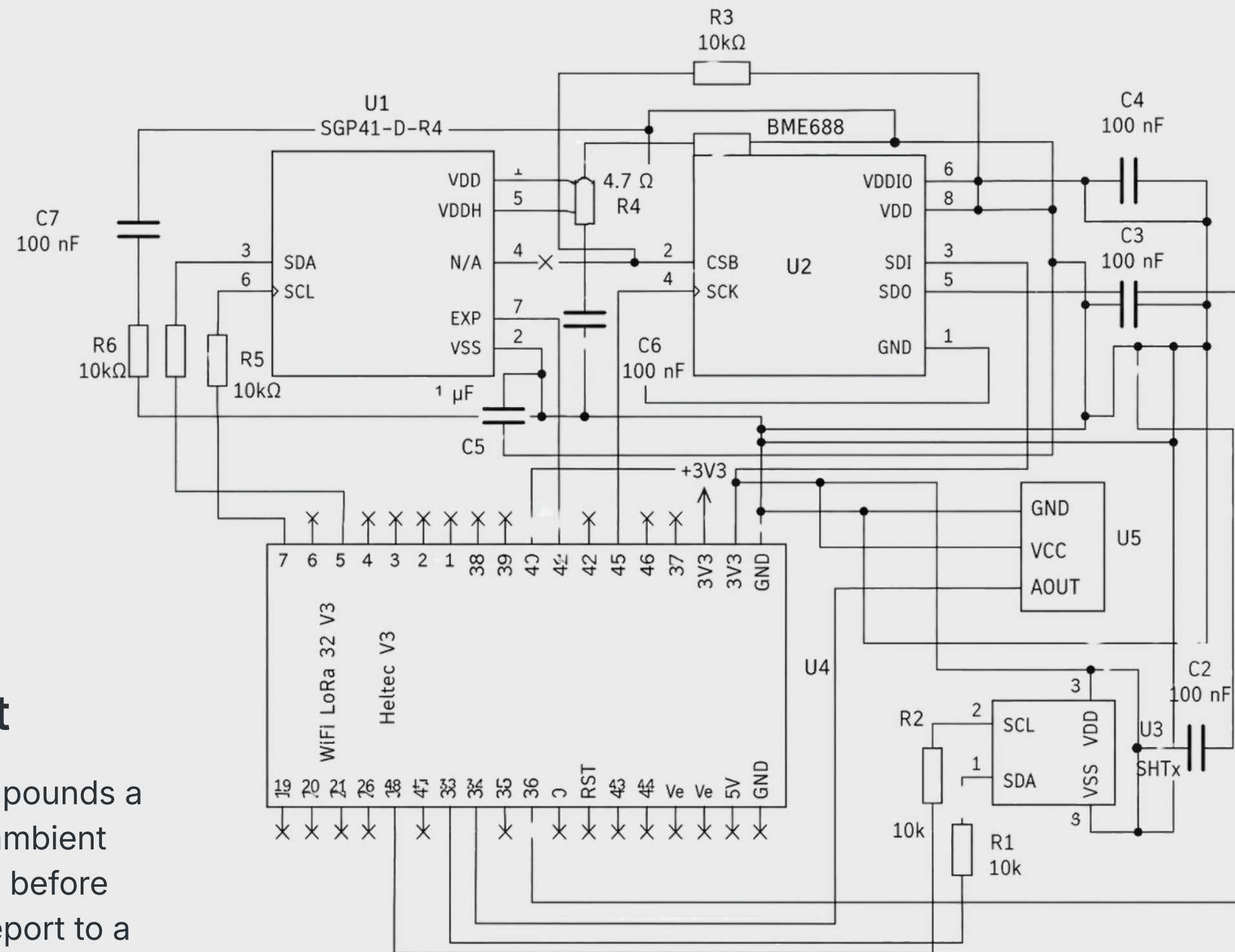
63M+ tons

Lost crops in America from the past 4 years

The Solution

About the Project

CropSentry reads the compounds a plant emits, plus soil and ambient conditions, to catch stress before symptoms show. Nodes report to a gateway to your phone.



Logic

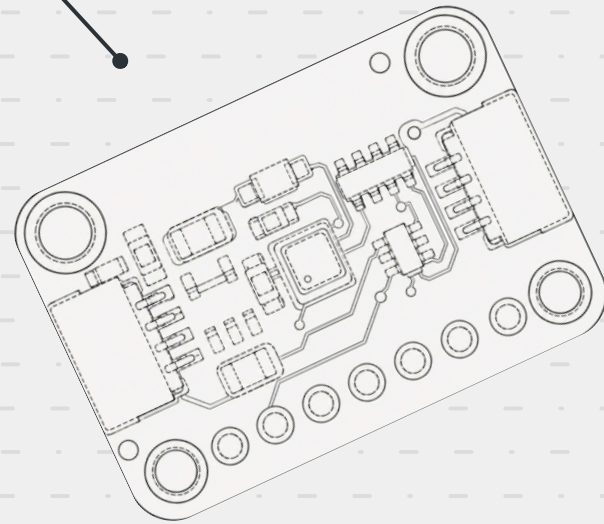
VOC (Volatile Organic Compound) sensors read the air that a plant emits. A temperature and humidity sensor provides the ambient baseline, isolating the plant's signal from the environment's.

Using solar power to charge the module, CropSentry is able to run fully autonomously and without excessive maintenance.

Hardware

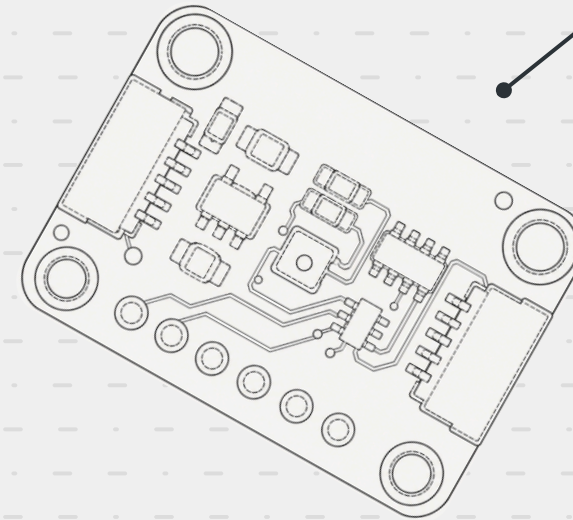
BME688

Checks for VOCs,
temperature, humidity, and
pressure



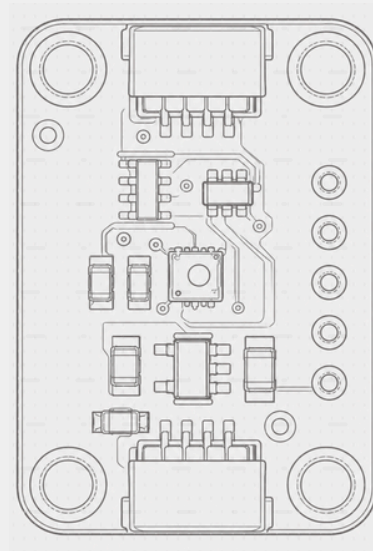
SHT45

Secondary temperature and
humidity sensor as an
ambient system



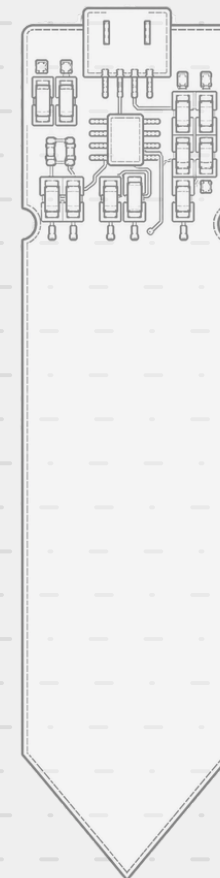
SGP41

Secondary VOC detection
system to ensure accurate
readings

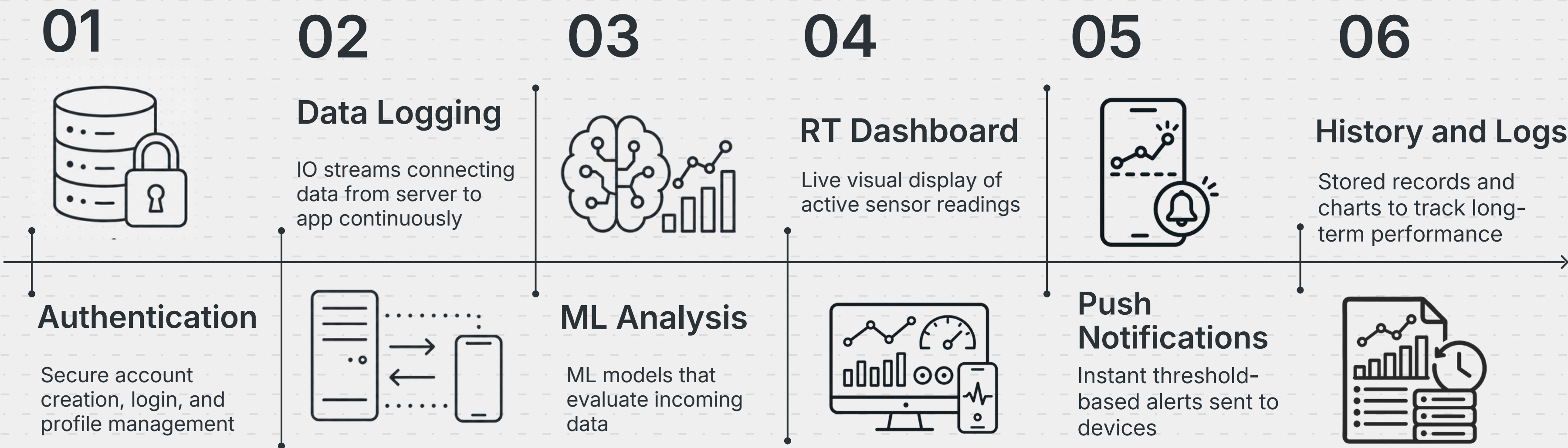


Soil Moisture Probe

Detects when water in the soil
is insufficient and sends an
alert to the user for watering



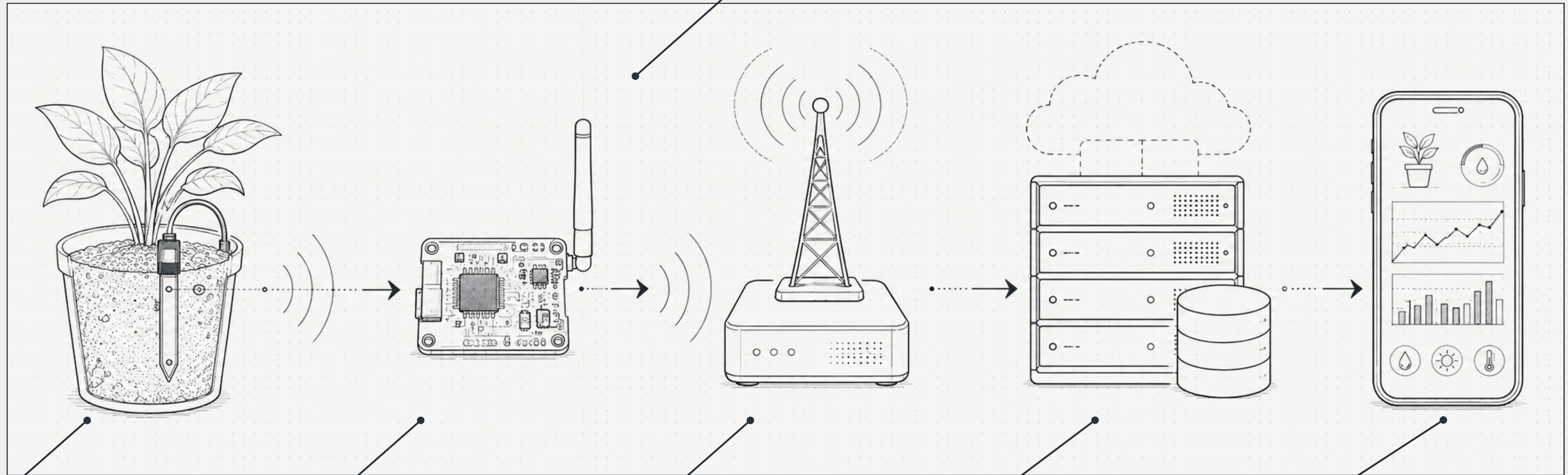
Software



System Architecture

Vision

To use multiple methods of data transfer to ensure range while also providing accurate readings to the user



I

Starts by collecting 56 data samples and sending one collection of info via radio signal

II

Takes radio signal and processes data through WiFi to be received through a server

III

Server processes information and sends details to the user's device through the app

IV

The app translates data and provides user-readable information about their plants

IV

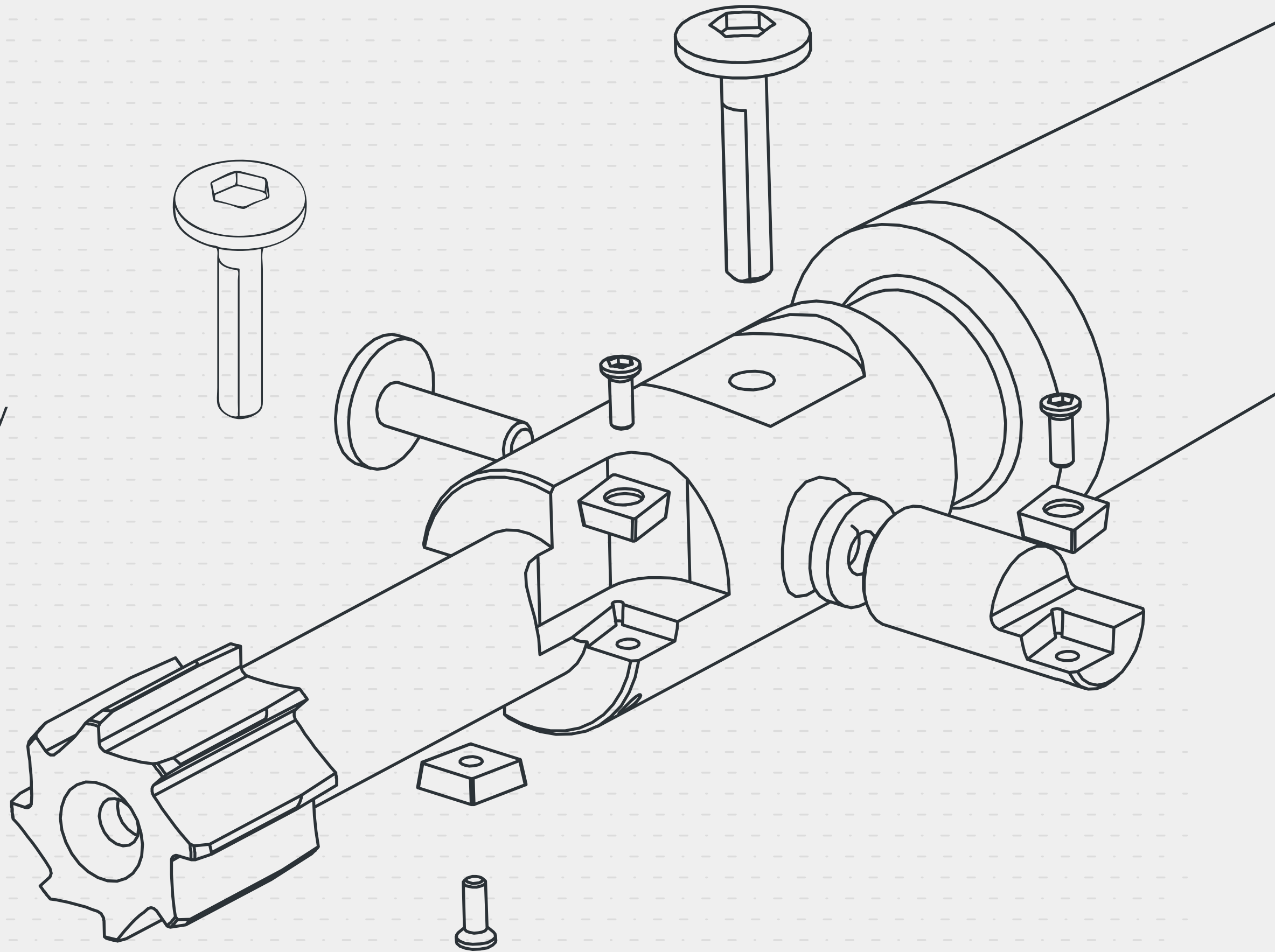
The user can virtually interact with their plant and get a full diagnosis regarding diseases and health

Potential



Q/A

Thank you for being a part of this presentation! Any questions are appreciated.



References

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